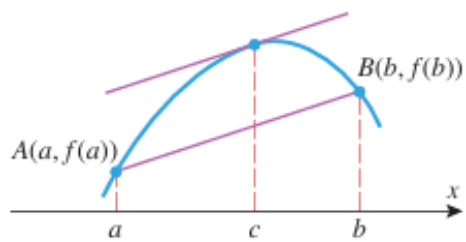
Lecture 9

Two Important Theorems

Mean Value Theorem:

Statement: Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). Then, there is at least one point c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Geometrically, this theorem states that between any two points $A(a, f(a))$ and $B(b, f(b))$ on the graph of a differentiable function f, there is at least one place where the tangent line to the graph is parallel to the secant line joining A and B.

Example: Verify the hypothesis of the Mean Value Theorem for the function $f(x) = \sqrt{4x - 3}$ over the interval [1,3], and find all values of c in the interval that satisfy the conclusion of the theorem.

Solution:

Continuity

$f(x) = \sqrt{4x - 3}$ is continuous if $\sqrt{4x - 3} \geq 0$

So, $\sqrt{4x - 3} \geq 0$

$\Rightarrow 4x - 3 \quad \geq 0$

$\Rightarrow 4x \quad \geq 3$

$\Rightarrow x \quad \geq \frac{3}{4}$

Here, $x \geq \frac{3}{4}$ is the condition for continuity.

$\therefore f(x)$ is continuous on [1,3].

<u>Differentiability</u>

$f(x) = \sqrt{4x-3}$

So, $f'(x) \quad = \dfrac{1}{2\sqrt{4x-3}} \cdot 4$

$\qquad\qquad = \dfrac{2}{\sqrt{4x-3}}$

Here, $\sqrt{4x-3} > 0$

$\Rightarrow 4x - 3 > 0$

$\Rightarrow x > \dfrac{3}{4}$

$\therefore f(x)$ is differentiable on $(1,3)$.

So, the hypotheses of Mean Value Theorem are satisfied.

Let, $c \in (1,3)$.

Here,

$f'(x) \quad = \dfrac{2}{\sqrt{4x-3}}$

$\therefore f'(c) \quad = \dfrac{2}{\sqrt{4c-3}}$

Now,

$f(x) = \sqrt{4x-3}$

$a = 1$

$b = 3$

$\therefore f(1) \quad = \sqrt{4(1)-3} = \sqrt{4-3} = 1$

and $f(3) \quad = \sqrt{4(3)-3} = \sqrt{12-3} = \sqrt{9} = 3$

Now, we know that,

$f'(c) = \dfrac{f(b) - f(a)}{b - a}$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = \frac{f(3)-f(1)}{3-1}$$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = \frac{3-1}{3-1}$$

$$\Rightarrow \frac{2}{\sqrt{4c-3}} = \frac{2}{2} = 1$$

$$\Rightarrow 2 = \sqrt{4c-3}$$

$$\Rightarrow 4c-3 = 4$$

$$\Rightarrow 4c = 4+3 = 7$$

$$\Rightarrow c = \frac{7}{4}$$

$$\therefore c = \frac{7}{4}$$

So, the function satisfies the Mean Value Theorem for $c = \frac{7}{4} \in (1,3)$. [Verified]

H.W. Verify the hypothesis of the Mean Value Theorem for the function $f(x) = 3 + 2x - x^2$ over the interval $[0,1]$, and find all values of c in the interval that satisfy the conclusion of the theorem.
Answer: $c = \frac{1}{2}$

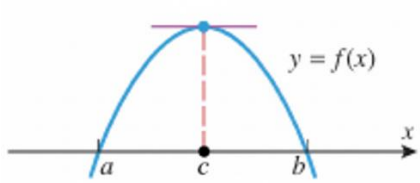
H.W. Exercise 4.8 (5-8) and Practice Sheet

Rolle's Theorem: (A Special Case of MVT)

Statement: Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). If

$$f(a) = 0 \quad \text{and} \quad f(b) = 0$$

then there is at least one point c in the interval (a, b) such that $f'(c) = 0$.



Geometrically, this theorem states the obvious fact that if the graph of a differentiable function intersects the x-axis at two points, a and b, then somewhere between a and b there must be at least one place where the tangent line is horizontal.

Example: Verify the hypothesis of Rolle's Theorem for the function $f(x) = 2x^3 + x^2 - 4x - 2$

in the interval $[-\sqrt{2}, \sqrt{2}]$, and find all values of c in the interval that satisfy the conclusion of the theorem.

Solution:

<u>Continuity</u>

$f(x) = 2x^3 + x^2 - 4x - 2$ is a polynomial.

Therefore, $f(x)$ is continuous on $\left[-\sqrt{2}, \sqrt{2}\right]$.

<u>Differentiability</u>

$f(x) \quad = 2x^3 + x^2 - 4x - 2$

$\therefore f'(x) \quad = 6x^2 + 2x - 4$

So, $f(x)$ is differentiable on $\left(-\sqrt{2}, \sqrt{2}\right)$.

Let, $c \in \left(-\sqrt{2}, \sqrt{2}\right)$.

Here,

$f'(x) \quad = 6x^2 + 2x - 4$

$\therefore f'(c) \quad = 6c^2 + 2c - 4$

$f(x) = 2x^3 + x^2 - 4x - 2$

$a \ = -\sqrt{2}$

$b \ = \sqrt{2}$

$$\begin{aligned} f(a) \quad &= f\left(-\sqrt{2}\right) = 2\left(-\sqrt{2}\right)^3 + \left(-\sqrt{2}\right)^2 - 4\left(-\sqrt{2}\right) - 2 = 0 \\ &= 0 \\ f(b) \quad &= f\left(\sqrt{2}\right) = 2\left(\sqrt{2}\right)^3 + \left(\sqrt{2}\right)^2 - 4\left(\sqrt{2}\right) - 2 \\ &= 0 \end{aligned}$$

$\therefore f(a) = f(b) = 0$

So, the hypothesis of Rolle's theorem is satisfied.

Now, $f'(c) = 0$

$\Rightarrow 6c^2 + 2c - 4 \quad = 0$

$\Rightarrow 3c^2 + c - 2 \quad = 0$

$$\Rightarrow c \quad = \frac{-1 \pm \sqrt{1^2 - 4 \times 3 \times (-2)}}{2 \times 3}$$

$$\Rightarrow c \quad = \frac{-1 \pm \sqrt{1 + 24}}{6}$$

$$\Rightarrow c \quad = \frac{-1 \pm \sqrt{25}}{6}$$

$\Rightarrow c \quad = \dfrac{-1 \pm 5}{6} = \dfrac{4}{6}, \dfrac{-6}{6}$

$\therefore c = \dfrac{2}{3}, -1$

Here, $c = \frac{2}{3} \in \left(-\sqrt{2}, \sqrt{2}\right)$ and $c = -1 \in \left(-\sqrt{2}, \sqrt{2}\right)$.

$\therefore$ The function satisfies Rolle's Theorem for $c = \dfrac{2}{3}, -1$

H.W. Verify Rolle's Theorem for the function $f(x) = x^2 + 5x - 6$ over $[-6,1]$.

Answer: $c = -\frac{5}{2} \in (-6,1)$

H.W. Exercise 4.8 (1-4) and Practice Sheet